

1 Graphene: Basic Properties

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1.1 Chemical Bonding and Ground-State Structure

Carbon is the sixth element in the periodic table. The carbon atom has six electrons, two of them in the $1s^2$ shell and four filling $2s$ and $2p$ states. The ground-state atomic configuration is $2s^2 2p^2$, with total spin $S = 1$, total orbital moment $L = 1$, and total angular momentum $J = 0$ (the ground-state multiplet 3P_0). The first excited state with $J = 1$, the 3P_1 multiplet, has energy $16.4 \text{ cm}^{-1} \approx 2 \text{ meV}$ [1] which gives an estimate of the spin-orbit coupling strength. The lowest energy state with $2s^1 2p^3$ configuration has excitation energy $33735.2 \text{ cm}^{-1} \approx 4.2 \text{ eV}$ [1], representing the promotion energy of a $2s$ electron to a $2p$ state. Therefore, we might expect carbon to be always divalent, due to the two $2p$ -electrons and the closed $2s$ shell. This conclusion is, however, wrong. Normally, carbon is tetravalent, due to the formation of hybridized sp states, according to the concept of “resonance” by Linus Pauling [2, 3].

When atoms form molecules or solids, the total energy decreases due to overlap of the electron wave functions at different sites and the formation of molecular orbitals (in molecules), or energy bands (in solids). This energy gain can be sufficient for the energy necessary to promote electrons from $2s$ to $2p$ states in the carbon atom. In non-ionic solids, there are two kinds of strong chemical bonding: covalent and metallic. In covalent bonds, the electrons are shared between neighboring atoms with a strong dependence of the cohesive energy on the angles between bonds, whereas in metallic bonding, the electrons are redistributed over the whole crystal in a more isotropic way. In diamond, four sp^3 covalent bonds with tetrahedral orientation are formed, with bond length $a \approx 1.54 \text{ \AA}$. Alternatively, three electrons can form sp^2 covalent bonds with planar minimum energy configuration and 120° angles, whereas one p -electron per atom forms a metallic bond. The prototype is the conjugated bond in the benzene molecule where the interatomic distances are $a \approx 1.42 \text{ \AA}$ shorter than in diamond. Graphite, the most common form of elemental carbon at ambient conditions, consists of flat layers with this type of covalent and metallic character. When isolated, a single layer is called *graphene*. In graphite, the graphene layers are weakly coupled by van der Waals interactions with an equilibrium interlayer distance $3.35 \text{ \AA}$ [4]. The minimum energy corresponds to Bernal stacking AB where even layers (B) are rotated on odd layers by 60° , as shown in Fig. 1.1. A slightly less favorable configuration is the rhombohedral graphite with three periodically repeated layers ABC, each rotated by $+60^\circ$ with respect to the adjacent ones. Structures with a random distributions of $+60^\circ$ and -60° rotations are called

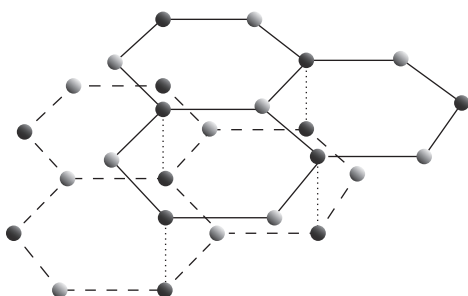


Fig. 1.1 Bernal stacking: carbon atoms belonging to sublattices *A* and *B* are shown as dark and light gray

turbostratic graphite. The AA stacking is energetically costly, and is not found in nature. The energetics of different types of stacking are discussed, e.g., in [5, 6]. Similar sp^2 -type bonding is also found in nanotubes and fullerenes [7]. Nanotubes can be thought of as graphene rolled at given angles and leading to a structure similar to the planar hexagonal structure of graphene but with slightly bent hexagons. Fullerenes are large molecules (C_{60} , C_{72} , ...) which can be considered as graphene pieces curved into objects with the topology of a sphere. For topological reasons (different Euler characteristics for sphere and cylinder), they cannot be built from hexagons only and require the introduction of 12 pentagons per molecule [7]. All these allotropes of elemental carbon, with the exception of diamond, are called graphitic forms.

As seen in the phase diagram shown in Fig. 1.2, at ambient conditions graphite is the stable structure, which is why it is much more common on the Earth's surface whereas at large pressures, like in the Earth's mantle, diamond is more favorable. The difference between graphite and diamond is very small but with a large barrier to be overcome for the transformation so that diamond exists as a metastable structure up to high temperature. In Fig. 1.3, we show how the energy change when pressure is applied perpendicularly to graphite layers leads to the transformation to diamond [8, 9]. The stability of the sp^2 bonding with respect to sp^3 for carbon is the reason why layers can be easily separated. In 2004, it was shown that single layers could be exfoliated and deposited on other substrates [10, 11] leading to the beginning of the “graphene boom” and opening the door to new two-dimensional (2D) materials. Other elements of the same fourth column of the periodic table strongly prefer sp^3 hybridization. For Si, Ge, and Sn, the diamond structure corresponds to the minimal energy. For Sn, this is the ground state at low temperature (grey tin) competing with the metallic phase (white tin) at higher temperatures ($T > 286$ K), whereas Pb is a metal with an fcc structure. Nevertheless, we can obtain 2D analogues of graphene for Si [12] and Ge [13], e.g. by epitaxial growth on the (111) surfaces of fcc metals. Exfoliation is of course impossible because, as said, there is not a bulk graphitic form for these elements. While for graphene in its ground state, the chemical bond is of pure sp^2 character and the structure is exactly planar, for silicene and germanene there is an essential sp^2 – sp^3 mixture of the bonding resulting in a buckled configuration. As shown in Fig. 1.4, due to the buckling in germanene the electronic density is strongly modulated in the

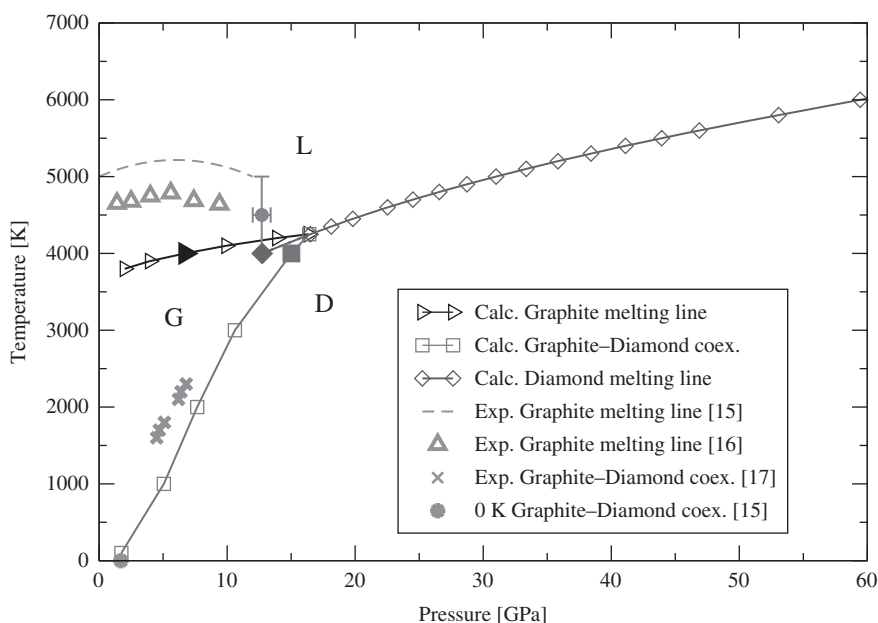


Fig. 1.2 Temperature–pressure phase diagram of carbon including the graphite, diamond, and liquid phase (adapted from [14]).

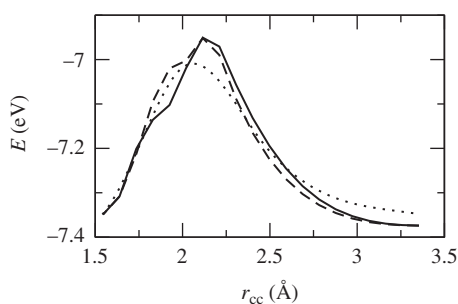


Fig. 1.3 Reaction path of the bulk diamond to graphite transformation as a function of the carbon–carbon distance perpendicular to the layers (adapted from [9]). Dotted line (DFT results from [8], dashed line [9], solid line [29]).

z direction, contrary to graphene. This results in particular in a much stronger spin–orbit coupling in silicene and germanene with potential relevance for spintronics [13].

1.2 Thermal (In)Stability of 2D Crystals

Many properties of graphene are studied for graphene deposited on a substrate, such as SiO_2 , hexagonal boron nitride (h-BN), and mica. Nevertheless, it is instructive to

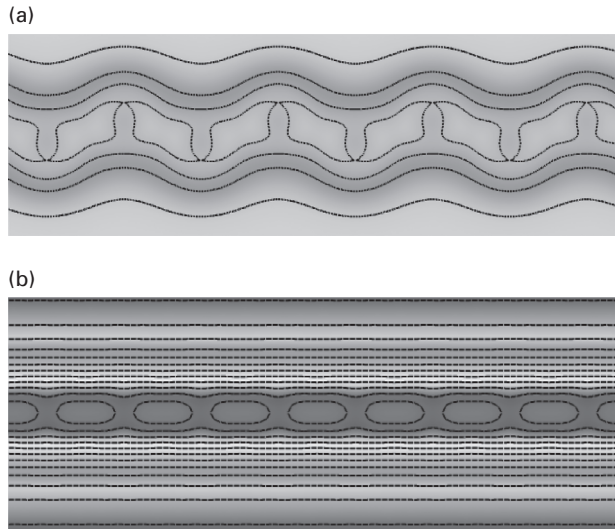


Fig. 1.4 Side view of charge-density distribution in germanene (a) and graphene (b) (courtesy A. N. Rudenko).

consider the properties of freely suspended graphene that can give general results for 2D structures embedded in 3D space. This is not only relevant for all other 2D crystals but also for the general long-standing problem of mechanics and statistical physics of thin membranes [18, 19, 20].

In crystals, the phonon spectrum always includes acoustic modes, with linear dispersion $\omega \propto q$ for $q \rightarrow 0$, where ω is the phonon frequency and q is the wavevector. As a result, in the harmonic approximation, at finite temperature the contribution of these modes to the mean square atomic displacement is divergent as $\langle u_q^2 \rangle \propto 1/q^2$. In 2D structures, this leads to divergence of the total mean square displacements as $\langle u^2 \rangle \propto \ln(L/a)$, where L is the sample size. This was first noticed in the 1930s by Peierls [21] and Landau [22] and led them to the conclusion that 2D crystals cannot be stable. Strictly speaking, this divergence only implies that the harmonic approximation, based on the assumption that thermal displacements are much smaller than interatomic distances, is not valid in 2D crystals. However, this statement on destruction of the long-range crystalline order in 2D crystals turned out to be correct and rigorously proved as a part of the Mermin–Wagner theorem [23]. Therefore, the statement that graphene is a 2D crystal requires some explanation and clarification.

If we embed 2D crystals in 3D space, the situation seems to be even worse. In fact, in this case, acoustic out-of-plane phonons turn out to have quadratic dispersion $\omega \propto q^2$ [24, 19, 20] leading to thermal instability of the flat state due to strong divergence of out-of-plane fluctuations $\langle h^2 \rangle \propto L^2$. Statistical mechanics of fluctuating membranes (for a review see [18, 19, 20]) show that an harmonic coupling of in-plane and out-of-plane modes is crucially important. This coupling can be described by scaling considerations similar to those used in the theory of critical phenomena. As a result, out-of-plane acoustic modes become less soft $q^2 \rightarrow q^{2-\eta}$ but in-plane modes become

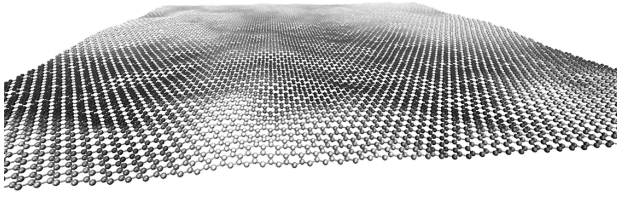


Fig. 1.5 Snapshot of graphene at room temperature as calculated by molecular dynamics simulations with the reactive potential LCBOPII [29] for a sample of 12,096 carbon atoms. The gray scale gives the amplitude of out-of-plane fluctuations, which, for this sample, ranges between +3 Å and −3 Å.

even softer $q \rightarrow q^{2-\eta}$. The best estimate of the parameter η from both atomistic Monte Carlo simulations and the functional renormalization group is 0.85 [20]. As a result, the membrane becomes flatter $\langle h^2 \rangle \propto L^{2-\eta}$ but the thermal instability of in-plane modes becomes even stronger than in the harmonic approximation $\langle u^2 \rangle \propto L^{2-2\eta}$. For this reason, long-range crystal order does not exist also for 2D crystals embedded in 3D space. The long-range crystal order is observable through the existence of Bragg peaks in the structure factor, which become infinitely sharp in the limit $L \rightarrow \infty$. For 2D crystals in 3D, these peaks have a finite broadening [19]. This broadening is small if $k_B T \ll \kappa$, where κ is the bending rigidity, and the centers of the broadened Bragg peaks still form the ideal reciprocal lattice. For graphene at room temperature, $k_B T / \kappa \approx 1/40$, thus justifying the use of the name crystal for graphene [20].

In turn, the behavior predicted by scaling theory $\langle h^2 \rangle \propto L^{2-\eta}$ means that at any finite temperature graphene has a rippled structure due to out-of-plane fluctuations. This is clearly seen in atomistic Monte Carlo simulations [25, 26], where ripples reversibly appear with temperature with amplitude following the above scaling with L . A typical snapshot for graphene at room temperature is shown in Fig. 1.5. Rippling of suspended graphene was experimentally observed by electron diffraction [27], although it is still under debate if these ripples are intrinsic due to dynamical thermal fluctuations or static, e.g., due to thermal compression [28].

1.3 Electronic Structure of Single-Layer Graphene

Graphene has the honeycomb crystal lattice shown in Fig. 1.6(a). The lattice vectors of the triangular Bravais lattice are

$$\vec{a}_1 = \frac{a}{2} (3, \sqrt{3}), \quad \vec{a}_2 = \frac{a}{2} (3, -\sqrt{3}), \quad (1.1)$$

where $a \approx 1.42$ Å is the nearest-neighbor distance. The unit cell contains two atoms belonging to two sublattices, *A* and *B*. Each atom of sublattice *A* is surrounded by three atoms of sublattice *B*, and vice versa. The vectors connecting each atom to the three nearest neighbors are

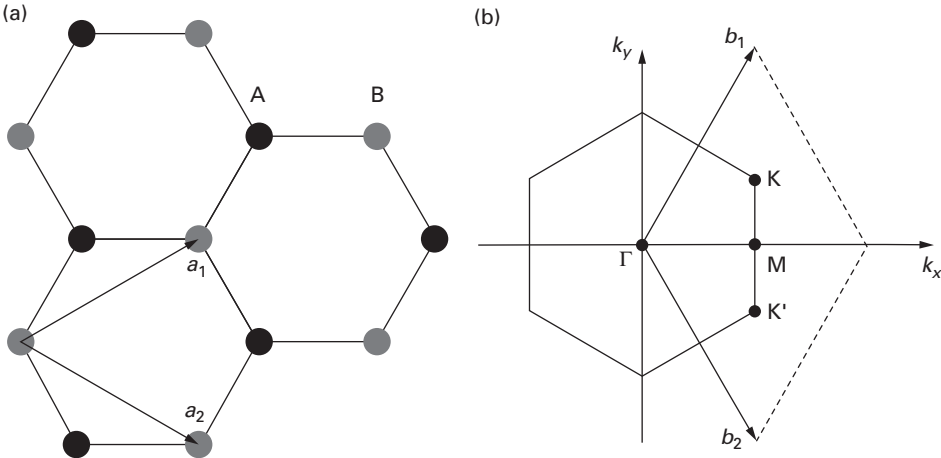


Fig. 1.6 (a) Honeycomb lattice with lattice vectors. Sublattices A and B are shown as black and gray. (b) Reciprocal lattice vectors and some special points in the Brillouin zone.

$$\vec{\delta}_1 = \frac{a}{2}(1, \sqrt{3}), \quad \vec{\delta}_2 = \frac{a}{2}(1, -\sqrt{3}), \quad \vec{\delta}_3 = a(-1, 0). \quad (1.2)$$

The reciprocal lattice of the triangular Bravais lattice is also triangular, with lattice vectors

$$\vec{b}_1 = \frac{2\pi}{3a}(1, \sqrt{3}), \quad \vec{b}_2 = \frac{2\pi}{3a}(1, -\sqrt{3}). \quad (1.3)$$

In Fig. 1.6(b), we show the Brillouin zone with the special high-symmetry points K , K' , and M

$$\vec{K}' = \left(\frac{2\pi}{3a}, \frac{2\pi}{3\sqrt{3}a}\right), \quad \vec{K} = \left(\frac{2\pi}{3a}, -\frac{2\pi}{3\sqrt{3}a}\right), \quad \vec{M} = \left(\frac{2\pi}{3a}, 0\right). \quad (1.4)$$

The electronic structure of graphene and graphite are discussed in detail in [30]. In Fig. 1.7, we show the graphene band structure. The sp^2 hybridized states (σ -states) form occupied and empty bands with a huge gap, whereas π -states form a single band with a conical self-crossing point at K (and by symmetry also at K'). The assignment to σ - and p_z -bands is done as in [31]. This band structure was obtained for the first time by Wallace in 1947 [32], by a simple tight-binding model, later developed by McClure [33] and Slonczewski and Weiss [34]. The Fermi energy of undoped graphene falls exactly at the crossing of the conical points, separating a completely filled valence band and an empty conduction band with no band gap in between. Therefore, at first sight, graphene might be classified as a *gapless semiconductor* [35]. Three-dimensional crystals, such as HgTe or α -Sn (gray tin), are known to be gapless semiconductors. What makes graphene unique is not the gapless state itself but the very unusual chiral nature of the electron states, which will be described later, as well as the high degree of electron–hole symmetry. As a result of subtle quantum relativistic effects (conductivity via evanescent modes of the Dirac operator), undoped graphene has a finite conductivity at zero

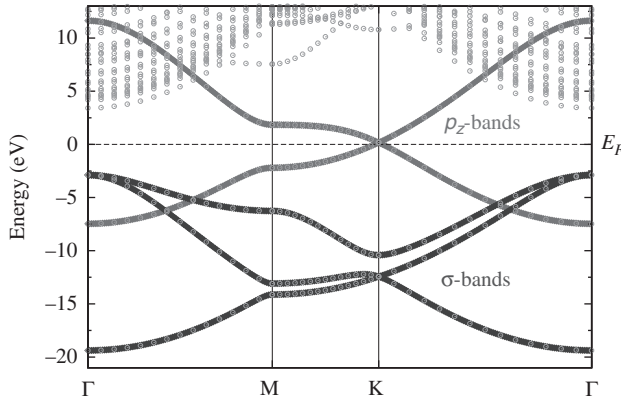


Fig. 1.7 Band structure of graphene (courtesy A. N. Rudenko).

temperature, despite the vanishing Fermi surface, becoming a *semimetal* rather than a gapless semiconductor [36, 19].

Following [32], we consider only the π -states and nearest-neighbor hopping terms (parameter t), so that there are only hopping processes from one sublattice to the other (*bipartite lattice*). The basis of the electronic states contains two π -states per unit cell belonging to atoms of sublattices A and B . The tight-binding Hamiltonian is therefore described by a 2×2 matrix

$$\hat{H}(\vec{k}) = \begin{pmatrix} 0 & tS(\vec{k}) \\ tS^*(\vec{k}) & 0 \end{pmatrix}, \quad (1.5)$$

where $\vec{k}$ is the wavevector and

$$S(\vec{k}) = \sum_{\vec{\delta}} e^{i\vec{k} \cdot \vec{\delta}} = 2 \exp\left(\frac{ik_x a}{2}\right) \cos\left(\frac{k_y a \sqrt{3}}{2}\right) + \exp(-ik_x a). \quad (1.6)$$

The resulting energy spectrum is

$$E(\vec{k}) = \pm |tS(\vec{k})| = \pm t \sqrt{3 + f(\vec{k})}, \quad (1.7)$$

where

$$f(\vec{k}) = 2 \cos(\sqrt{3}k_y a) + 4 \cos\left(\frac{\sqrt{3}}{2}k_y a\right) \cos\left(\frac{3}{2}k_x a\right). \quad (1.8)$$

Since $S(\vec{K}) = S(\vec{K}') = 0$, the two bands cross at these points. By expanding the Hamiltonian near these points and eliminating a constant phase factor by a proper gauge choice, we find

$$\hat{H}_K(\vec{q}) \approx \frac{3at}{2} \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix}, \quad (1.9)$$

where $\vec{q} = \vec{k} - \vec{K}$, $\hat{H}_{K'}(\vec{q}) = [\hat{H}_{K'}(\vec{q}')]^T$ with $\vec{q}' = \vec{k} - \vec{K}'$ and the superscript T indicates the transpose of the matrix. We can interpret the prefactor in the right-hand side of Eq. (1.9) as $\hbar v$, where

$$v = \frac{3at}{2\hbar} \quad (1.10)$$

is the electron velocity at the conical points. Experimentally, $v \approx c/300 \approx 10^6$ m/s where c is the velocity of light [37, 38].

Taking into account the next-nearest-neighbor hopping t' we find, instead of Eq. (1.7)

$$E(\vec{k}) = \pm t|S(\vec{k})| + t'f(\vec{k}) = \pm t\sqrt{3 + f(\vec{k})} + t'f(\vec{k}). \quad (1.11)$$

The term in t' breaks the electron–hole symmetry shifting the conical point from $E = 0$ to $E = -3t'$ but it does not change the behavior of the Hamiltonian near the conical points. Experimentally, t' is negative (the sign of t is arbitrary since it can be eliminated by a gauge transformation of the wave function in one of the sublattices) and $t' \approx -0.1|t|$ [39]. The existence of the conical points in graphene is a consequence of symmetry [34], and therefore it does not depend on any specific assumptions on the number of bands, hopping parameters etc. The protection of this feature relies on two discrete symmetries: time reversal, and spatial inversion [40]. The latter assumes equivalence of the sublattices.

The points K and $-K'$ are equivalent because they differ by the reciprocal lattice vector $\vec{b} = \vec{b}_1 + \vec{b}_2$. This equivalence becomes evident using a representation such as the one shown in Fig. 1.8, with a larger unit cell in reciprocal space and six conical points.

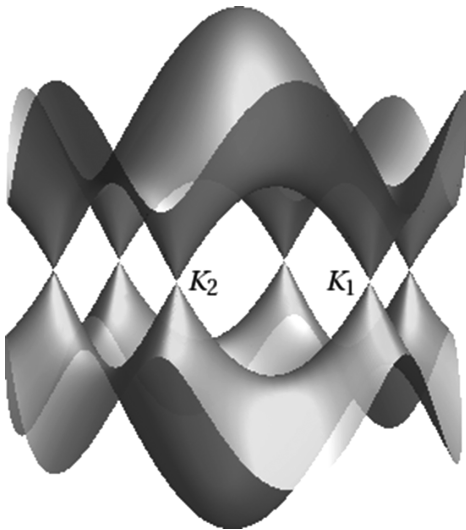


Fig. 1.8 Electron energy spectrum of graphene in the nearest-neighbor approximation (Eq. (1.7)) in a larger unit cell in reciprocal space with six conical points. The points K_1 and K_2 are equivalent to K and $-K'$.

For any realistic doping, the Fermi energy in graphene is close to the energy at the conical point, $|E_F| \ll |t|$. To describe electron and hole states in this regime, we need to expand the effective Hamiltonian near the special points K and K' and replace $q_x \rightarrow -i\frac{\partial}{\partial x}$, $q_y \rightarrow -i\frac{\partial}{\partial y}$ which corresponds to the effective mass approximation, or $\vec{k} \cdot \vec{p}$ perturbation theory [41]. From Eq. (1.9), we obtain

$$\hat{H}_K = -i\hbar v \vec{\sigma} \cdot \nabla, \quad (1.12)$$

where

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1.13)$$

are the Pauli matrices (only the x - and y -components enter Eq. (1.12)). A complete low-energy Hamiltonian consists of 4×4 matrices taking into account both sublattices and two conical points (in terms of semiconductor physics, two valleys).

On the basis that

$$\Psi = \begin{pmatrix} \psi_{KA} \\ \psi_{KB} \\ \psi_{K'A} \\ \psi_{K'B} \end{pmatrix}, \quad (1.14)$$

where, e.g., ψ_{KA} labels the component of the electron wave function corresponding to valley K and sublattice A , the Hamiltonian is a 2×2 block super matrix

$$\hat{H} = \begin{pmatrix} \hat{H}_K & 0 \\ 0 & \hat{H}_{K'} \end{pmatrix}. \quad (1.15)$$

A more symmetric form of the Hamiltonian is obtained using another basis

$$\Psi = \begin{pmatrix} \psi_{KA} \\ \psi_{KB} \\ \psi_{K'B} \\ -\psi_{K'A} \end{pmatrix} \quad (1.16)$$

so that the Hamiltonian (Eq. (1.15)) becomes

$$\hat{H} = -i\hbar v \tau_0 \otimes \vec{\sigma} \cdot \nabla, \quad (1.17)$$

where τ_0 is the unit matrix in valley indices. We will use different notations for the same Pauli matrices acting on different indices, namely $\vec{\sigma}$ in the sublattice space and $\vec{\tau}$ in the valley space.

For ideal graphene, the valleys are decoupled. In the presence of inhomogeneities, which are smooth at the atomic scale (external electric and magnetic fields, disorder, etc.), the valleys remain independent since the Fourier component of the external potential with the Umklapp wavevector $\vec{b}$ is very small, and intervalley scattering is improbable. However, we should keep in mind that any sharp (atomic-scale) inhomogeneity, like boundaries or vacancies, would mix the states from different valleys.

The Hamiltonian (Eq. (1.12)) is a 2D analogue of the Dirac Hamiltonian for massless fermions [42]. Instead of the velocity of light c , there is a parameter $v \approx c/300$. The internal degree of freedom, which is just spin for “true” Dirac fermions, is the sublattice index in the case of graphene. Dirac “spinors” consist here of the components giving the distribution of electrons in sublattices A and B . We will call this quantum number *pseudospin*, so that pseudospin “up” means sublattice A and pseudospin “down” means sublattice B . Beside the pseudospin, there are two more internal degrees of freedom, the valley label (sometimes called isospin) and the real spin, so that the most general low-energy Hamiltonian of electrons in graphene is an 8×8 matrix.

Spin–orbit coupling leads to a mixture of pseudospin and real spin and to gap opening [43]. However, the value of the gap is supposed to be very small, of the order of $10^{-2}K$ for pristine graphene [44]. The reason is not only the lightness of carbon atoms but also the orientation of the orbital moments of π -states perpendicular to the graphene plane.

For the case of “true” Dirac fermions in 3D space, the Hamiltonian is a 4×4 matrix, due to two projections of the spin and two values of the charge degree of freedom (particle versus antiparticle). For the 2D case, separation in particle and antiparticle is not independent of the pseudospin direction. Electrons and holes in graphene are just linear combinations of the states from the sublattices A and B . The 2×2 matrix $\hbar v \vec{\sigma} \cdot \vec{k}$ (the result of the action of the Hamiltonian (Eq. (1.12)) on a plane wave with wavevector $\vec{k}$) is diagonalized by the unitary transformation

$$\hat{U}_{\vec{k}} = \frac{1}{\sqrt{2}}(1 + i\vec{m}_{\vec{k}} \cdot \vec{\sigma}), \quad (1.18)$$

where $\vec{m}_{\vec{k}} = (\cos \phi_{\vec{k}}, -\sin \phi_{\vec{k}})$ and $\phi_{\vec{k}}$ is the polar angle of the vector $\vec{k}$. The eigenfunction

$$\psi_{e,h}^{(K)}(\vec{k}) = \frac{1}{\sqrt{2}} \begin{pmatrix} \exp(-i\phi_{\vec{k}}/2) \\ \pm \exp(i\phi_{\vec{k}}/2) \end{pmatrix} \quad (1.19)$$

corresponds to electron (e) and hole (h) states, with energies

$$E_{e,h} = \pm \hbar v k. \quad (1.20)$$

For the valley K' , the corresponding states (in the basis (Eq. (1.14)) are

$$\psi_{e,h}^{(K')}(\vec{k}) = \frac{1}{\sqrt{2}} \begin{pmatrix} \exp(i\phi_{\vec{k}}/2) \\ \pm \exp(-i\phi_{\vec{k}}/2) \end{pmatrix}. \quad (1.21)$$

For the electron (hole) states, by definition

$$\frac{(\vec{k} \cdot \vec{\sigma})}{k} \psi_{e,h} = \pm \psi_{e,h}. \quad (1.22)$$

This means that the electrons (holes) in graphene have a definite pseudospin direction, namely, parallel (antiparallel) to the direction of motion. Thus, these states are *chiral*

(*helical*), as it should be for massless Dirac fermions [42]. This is of crucial importance for “relativistic” effects, such as Klein tunneling [45, 19], which will be considered briefly later.

The Dirac model for electrons in graphene results from the lowest-order expansion of the tight-binding Hamiltonian (Eq. (1.12)) near the conical points. Taking into account the next, quadratic, term, we find (in the basis (Eq. (1.16)), instead of the Hamiltonian (Eq. (1.17))

$$\hat{H} = \hbar v \tau_0 \otimes \vec{\sigma} \cdot \vec{k} + \mu \tau_z \otimes \left[2\sigma_y k_x k_y - \sigma_x (k_x^2 - k_y^2) \right], \quad (1.23)$$

where $\mu = 3a^2 t / 8$. The additional term in Eq. (1.23) corresponds to a *trigonal warping* [46]. Diagonalization of the Hamiltonian (Eq. (1.23)) gives the spectrum $E_{e,h}(\vec{k}) = \pm \varepsilon(\vec{k})$, where

$$\varepsilon^2(\vec{k}) = \hbar^2 v^2 k^2 \mp 2\hbar v \mu k^3 \cos(3\phi_{\vec{k}}) + \mu^2 k^4. \quad (1.24)$$

The signs $\mp$ in Eq. (1.24) correspond to valleys K and K' . The dispersion is no more isotropic but has a three-fold symmetry.

1.4 Electronic Structure of Bilayer Graphene

The process of exfoliation leads often to samples made of more than one graphene layer. Bilayer graphene [47] is especially interesting. To describe a bilayer, we need to extend the tight-binding model to consider interlayer terms [33, 34, 48].

The crystal structure of bilayer graphene with Bernal stacking is shown in Fig. 1.9. The atoms in the sublattices A of the two layers lie exactly on top of each other, with a finite hopping parameter γ_1 between them, whereas there are no significant hopping processes between the sublattices B of the two layers. Data on the electronic structure of graphite [49] indicate that the parameter $\gamma_1 \equiv t_{\perp}$ is about 0.4 eV, an order of magnitude smaller than the nearest-neighbor in-plane hopping parameter $\gamma_0 = t$. The simplest model, which takes into account only these processes, is described by the Hamiltonian

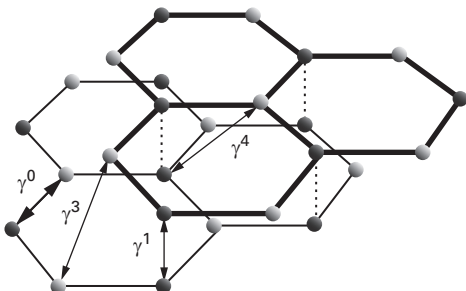


Fig. 1.9 Crystal structure of bilayer graphene, hopping parameters are shown.

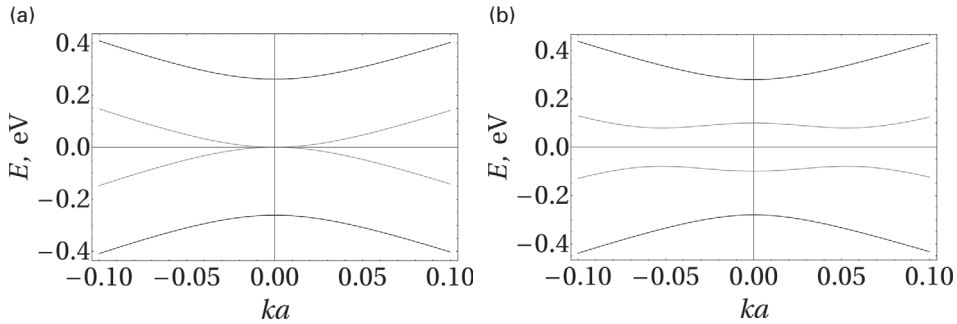


Fig. 1.10 (a) Electronic structure of bilayer graphene within the model Eq. (1.25). (b) The same, for the model Eq. (1.32).

$$\hat{H}(\vec{k}) = \begin{pmatrix} 0 & tS(\vec{k}) & t_{\perp} & 0 \\ tS^*(\vec{k}) & 0 & 0 & 0 \\ t_{\perp} & 0 & 0 & tS^*(\vec{k}) \\ 0 & 0 & tS(\vec{k}) & 0 \end{pmatrix} \quad (1.25)$$

with $S(\vec{k})$ from Eq. (1.6). The basis states are ordered as first layer, sublattice A ; first layer, sublattice B ; second layer, sublattice A ; second layer, sublattice B .

The matrix (Eq. (1.25)) can be easily diagonalized, with four eigenvalues

$$E_i(\vec{k}) = \pm \frac{1}{2}t_{\perp} \pm \sqrt{\frac{1}{4}t_{\perp}^2 + t^2|S(\vec{k})|^2} \quad (1.26)$$

with two independent $\pm$ signs. The spectrum is shown in Fig. 1.10. It contains two degenerate bands at the points K and K' with parabolic dispersion given by

$$E_{1,2}(\vec{k}) \approx \pm \frac{t^2|S(\vec{k})|^2}{t_{\perp}} \approx \pm \frac{\hbar^2 q^2}{2m^*}, \quad (1.27)$$

where the effective mass $m^* = |t_{\perp}|/(2v^2)$. Experimentally, $m^* \approx 0.028m_e$, where m_e is the mass of a free electron [50].

If we neglect intervalley scattering, we can construct the effective Hamiltonian describing only the two touching parabolic bands (Eq. (1.27)) by replacing $\hbar q_x$ and $\hbar q_y$ with operators $\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$ and $\hat{p}_y = -i\hbar \frac{\partial}{\partial y}$. The result reads [47, 48]

$$\hat{H}_K = \frac{1}{2m^*} \begin{pmatrix} 0 & (\hat{p}_x - i\hat{p}_y)^2 \\ (\hat{p}_x + i\hat{p}_y)^2 & 0 \end{pmatrix}. \quad (1.28)$$

The eigenstates of this Hamiltonian have very special chiral properties [47], resulting in a special Landau quantization, special scattering, etc. [19]. Electron and hole states corresponding to the energies

$$E_{e,h} = \pm \frac{\hbar^2 k^2}{2m^*} \quad (1.29)$$

have a form similar to Eq. (1.19), with the replacement $\phi_{\vec{k}} \rightarrow 2\phi_{\vec{k}}$

$$\psi_{e,h}^{(K)}(\vec{k}) = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\phi_{\vec{k}}} \\ \pm e^{i\phi_{\vec{k}}} \end{pmatrix}. \quad (1.30)$$

Instead of Eq. (1.22) the eigenfunctions Eq. (1.30) are characterized by the property

$$\frac{(\vec{k} \cdot \vec{\sigma})^2}{k^2} \psi_{e,h} = \pm \psi_{e,h}. \quad (1.31)$$

By applying a voltage V perpendicular to the carbon planes, we can open a gap in the energy spectrum [48, 51]. In this case, instead of the Hamiltonian (Eq. (1.25)), we have

$$H(\vec{k}) = \begin{pmatrix} V/2 & tS(\vec{k}) & t_{\perp} & 0 \\ tS^*(\vec{k}) & V/2 & 0 & 0 \\ t_{\perp} & 0 & -V/2 & tS^*(\vec{k}) \\ 0 & 0 & tS(\vec{k}) & -V/2 \end{pmatrix} \quad (1.32)$$

and, instead of eigenvalues Eq. (1.26), one has

$$E_i^2(\vec{k}) = t^2 |S(\vec{k})|^2 + \frac{t_{\perp}^2}{2} + \frac{V^2}{4} \pm \sqrt{\frac{t_{\perp}^4}{4} + (t_{\perp}^2 + V^2)t^2 |S(\vec{k})|^2}. \quad (1.33)$$

For the two low-lying bands in the vicinity of the K (or K') point, the spectrum has the “Mexican hat” dispersion

$$E(\vec{k}) \approx \pm \left(\frac{V}{2} - \frac{V\hbar^2 v^2}{t_{\perp}^2} k^2 + \frac{\hbar^4 v^4}{t_{\perp}^2 V} k^4 \right), \quad (1.34)$$

where we assume, for simplicity, $\hbar v k \ll V \ll |t_{\perp}|$. The expression in parenthesis has a maximum at $k = 0$ and a minimum at $k = \frac{V}{\sqrt{2}\hbar v}$ (see Fig. 1.10(b)). The possibility to tune the gap in bilayer graphene by an electric field was experimentally confirmed [51, 52].

Consider now the effect of hopping processes at larger distances, namely, between B sublattices ($\gamma_3 \approx 0.3$ eV [49]). Higher-order terms, such as $\gamma_4 \approx 0.04$ eV, are supposed to be negligible. The term γ_3 leads to a qualitative change in the spectrum near the K (K') point. The effective Hamiltonian is modified by γ_3 terms as [48]

$$\hat{H}_K = \begin{pmatrix} 0 & \frac{(\hat{p}_x - i\hat{p}_y)^2}{2m^*} + \frac{3\gamma_3 a}{\hbar} (\hat{p}_x + i\hat{p}_y) \\ \frac{(\hat{p}_x - i\hat{p}_y)^2}{2m^*} + \frac{3\gamma_3 a}{\hbar} (\hat{p}_x - i\hat{p}_y) & 0 \end{pmatrix} \quad (1.35)$$



Fig. 1.11 Effect of hopping parameter γ_3 on the band structure of bilayer graphene.

with the energy spectrum

$$E^2(\vec{k}) \approx (3\gamma_3 a)^2 k^2 + \frac{3\gamma_3 a \hbar^2 k^3}{m^*} \cos\left(3\phi_{\vec{k}}\right) + \left(\frac{\hbar^2 k^2}{2m^*}\right)^2. \tag{1.36}$$

This means that at small enough wavevectors

$$ka \leq \left|\frac{\gamma_3 \gamma_1}{\gamma_0^2}\right| \sim 10^{-2} \tag{1.37}$$

the parabolic dispersion law (Eq. (1.29)) is replaced by the linear one with four conical touching points instead of one parabolic touching point (see Fig. 1.11).

1.5 Graphene as a Bridge between Condensed Matter and High-Energy Physics

The many exceptional properties of graphene discussed in this chapter make it interesting in many respects and it will therefore be the subject of following chapters. Here we will underline the relevance of graphene for fundamental physics, beyond condensed matter. The key point is that charge carriers in graphene can be described as massless Dirac fermions. In particle physics, to reach this ultra-relativistic regime we need energies much higher than the particle rest energy whereas in graphene one can reach it by default. Moreover, to open a gap in the spectrum, which corresponds to the rest energy in particle physics, some efforts are needed, like chemical functionalization [53], combination of strain with electric gating [54], size quantization in nanoribbons [55]. Massless Dirac fermions appear also in other situations in condensed matter physics, for example in superfluid ^3He [56] and on the surface of topological insulators [57], but in

graphene the Dirac regime describes the energy spectrum in an particularly broad range up to a few eV. Since in graphene there are two Dirac cones instead of one as in topological insulators, the Dirac physics can be in principle destroyed by intervalley scattering processes due to atomically sharp defects. However, since graphene has one of the strongest chemical bonds in nature, such defects (vacancies, dislocations, Stone–Wales defects) have high-formation energy [29] and rarely occur spontaneously. Therefore, the observation of phenomena related to the Dirac nature of the charge carriers is not very challenging. An important difference between Dirac fermions and charge carriers in graphene is that the light velocity c is replaced by the Fermi velocity v which is 300 times smaller. As a result, instead of the fine structure constant $\alpha = e^2/\hbar c \approx 1/137$ determining the strength of interaction in quantum electrodynamics, in graphene there is a much stronger interaction with adimensional coupling constant $e^2/\hbar v \approx 2.2$. For this reason, some quantum relativistic effects, which are difficult to observe in particle physics, become accessible in the physics of graphene. Probably the best example is the relativistic atomic collapse which is supposed to happen for the elements with $Z > 170$, determining the natural limit of the periodic table [58] which is still far from being reached. For graphene, an analogous phenomenon was predicted [59, 60, 61] and experimentally observed [62]. Funnily, the true fine structure constant also plays a role in graphene, determining its optical transmission with absorption coefficient $\eta = \pi\alpha \approx 2.3\%$ per graphene layer [63].

Probably the most relevant quantum relativistic phenomenon for the physics and applications of graphene is Klein tunneling [45] which was also experimentally observed [64, 65]. This is a property of relativistic quantum particles, to be transmitted across a potential barrier of any height and thickness. This property is a direct consequence of chirality (Eq. (1.22)). On the one hand, Klein tunneling does not allow to make a graphene transistor based just on p–n junctions as in semiconductors because we cannot lock the junction. On the other hand, it is the Klein tunneling that protects the mobility of charge carriers in graphene, making the unavoidable presence of electron–hole puddles [66] not so detrimental to charge transport.

A detailed discussion of the relation of the physics of graphene to high-energy physics can be found in [19].

1.6 References

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